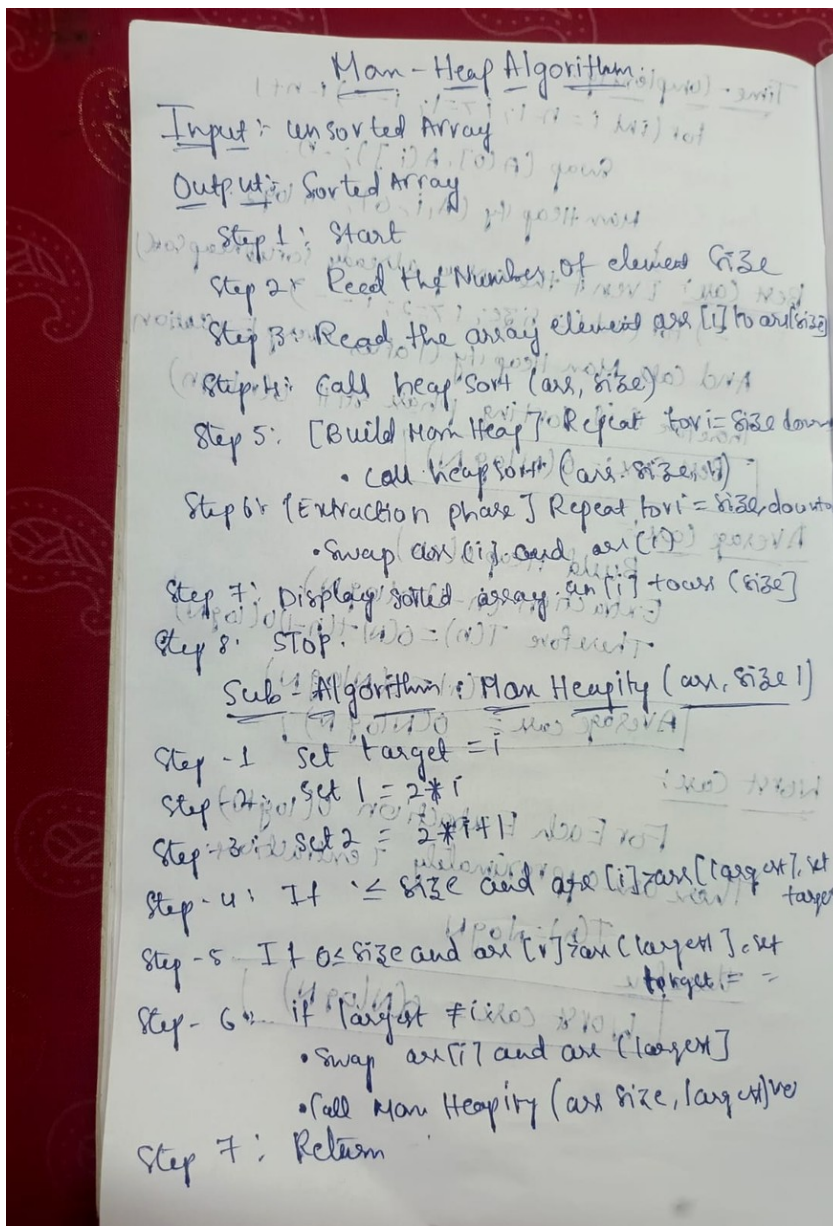


Practical: 3

Implementation of max-heap sort algorithm

Aim: To Implement the max-heap sort algorithm

Algorithm:



Program:

```
#include <iostream>
using namespace std;

class HeapSort
{
public:
    int arr[50], size;

    void inputArray()
    {
        cout << "Enter size of array: ";
        cin >> size;

        cout << "Enter " << size << " elements: ";
        for (int i = 0; i < size; i++)
        {
            cin >> arr[i];
        }
    }

    void MaxHeapify(int arr[], int size, int i)
    {
        int largest = i;
        int l = (2 * i) + 1;
        int r = (2 * i) + 2;

        if(l < size && arr[l] > arr[largest]){
            largest = l;
        }

        if(r < size && arr[r] > arr[largest]){
            largest = r;
        }

        if(largest != i)
        {
            swap(arr[largest], arr[i]);
            MaxHeapify(arr, size, largest);
        }
    }

    void heapSort()
    {
        for(int i = size/2 - 1; i >= 0; i--)
        {

```

```
        MaxHeapify(arr, size, i);
    }

    for(int i = size - 1; i >=1; i--)
    {
        swap(arr[0], arr[i]);
        MaxHeapify(arr, i, 0);
    }
}

void printArray()
{
    cout << "Sorted array: ";
    for (int i = 0; i < size; i++)
    {
        cout << arr[i] << " ";
    }
    cout << endl;
}

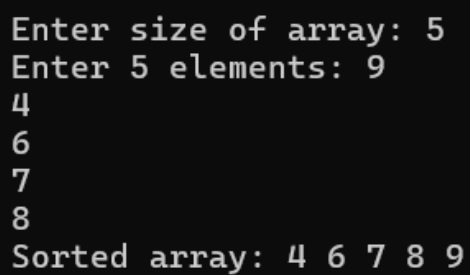
};

int main()
{
    HeapSort obj;

    obj.inputArray();
    obj.heapSort();
    obj.printArray();

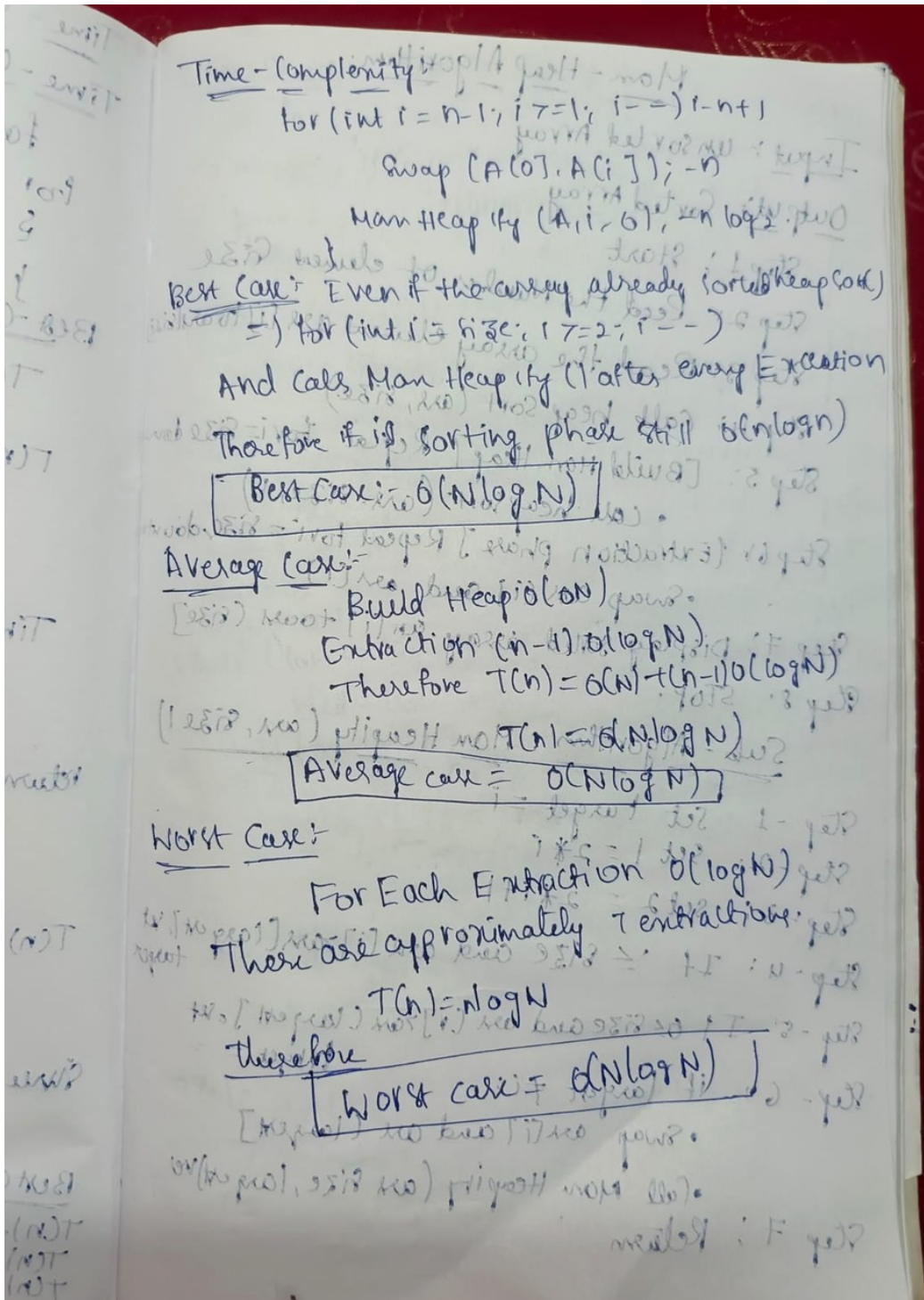
    return 0;
}
```

Output:



```
Enter size of array: 5
Enter 5 elements: 9
4
6
7
8
Sorted array: 4 6 7 8 9
-----
```

Time Complexity:



Conclusion:

Hence, the Max-Heap Sort algorithm was successfully implemented and executed. It was observed that Heap Sort gives a time complexity of $O(n \log n)$ in the best, average, and worst cases. It is an efficient sorting technique and works well for sorting large sets of data.

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